

CSE-370 Database Systems

Quiz-2 Set-A

ID, Name and section :

Time: 25 minutes

Q1. Consider a GAME database in which data is recorded about the video game industry. The data requirements are summarized as follows:

- Game studios are identified by name and each has an address. A game studio develops one or more games.
- Each game is identified by its title and year of release. Each game is developed by a game studio. Some games have memorable quotes, each of which is spoken by a particular character in the game.
- A game is directed by one or more game directors. Each director is identified by name and date of birth and can direct multiple games. It is possible for a game director to also be a voice actor, including for a game they direct.
- Each game features characters. Each character is voiced by a particular voice actor. Each voice actor is identified by name and date of birth and may work on multiple games.

Draw an EER diagram for the GAME database. **[10 Marks]**

Q2.a. The schema of a hospital database is shown below. For each table below identify primary key and foreign keys (if any). For a foreign key, write down which table the key refers to. **[5 marks]**

Patient

pID	name
1	John Doe
2	Jane Smith

Doctor

dID	name
1	Dr. Adams
2	Dr. Baker

Test

tID	testName
1	ECG
2	X-ray

Appointment

pID	dID	time
1	1	9pm 11-03-25
2	2	9pm 11-03-25

PrescribedTestsInAppointment

pID	dID	time	tID
1	1	9pm 11-03-25	1
1	1	9pm 11-03-25	2

Q2.b. Based on your answer of Q2.a, write down the relational integrity constraints each of the operations below violate (if any). Consider each operation to be independent. **[5 marks]**

1. Insert <1,1,10pm 11-03-25> in the Appointment Table
2. Insert <1,1,11pm 11-03-25, null> in the PrescribedTestsInAppointment Table
3. Delete the second row from the PrescribedTestsInAppointment Table.